

Calculus 2

4 Lecture 4

Product of Series

Definition 4.1. Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two series. The *Cauchy product* of these two series is defined as follows:

$$\left(\sum_{k=1}^{\infty} a_k \right) \left(\sum_{k=1}^{\infty} b_k \right) \stackrel{\text{def}}{=} \sum_{k=1}^{\infty} c_k, \quad c_k = a_1 b_k + a_2 b_{k-1} + \cdots + a_k b_1.$$

For example,

$$c_1 = a_1 b_1, \quad c_2 = a_1 b_2 + a_2 b_1, \quad c_3 = a_1 b_3 + a_2 b_2 + a_3 b_1.$$

Question. Is it true that

$$\left(\sum_{k=1}^{\infty} a_k = a \in \mathbb{R} \quad \text{and} \quad \sum_{k=1}^{\infty} b_k = b \in \mathbb{R} \right) \implies \left(\sum_{k=1}^{\infty} c_k = ab \right),$$

where $c_k = a_1 b_k + \cdots + a_k b_1$? In other words, if both series converge, must their Cauchy product converge to the product of their sums?

The following example shows that the answer is **no**.

Example 4.1. Let

$$a_k = b_k = \frac{(-1)^{k+1}}{\sqrt{k}}.$$

Then the series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ are conditionally convergent, due to [Claim 3.8](#) (alternating p -series). The terms of their Cauchy product are

$$\begin{aligned} c_k &= \frac{(-1)^{1+1}}{\sqrt{1}} \frac{(-1)^{k+1}}{\sqrt{k}} + \cdots + \frac{(-1)^{i+1}}{\sqrt{i}} \frac{(-1)^{k-i+2}}{\sqrt{k-i+1}} + \cdots + \frac{(-1)^{k+1}}{\sqrt{k}} \frac{(-1)^{1+1}}{\sqrt{1}} \\ &= (-1)^{k+3} \sum_{i=1}^k \frac{1}{\sqrt{i} \sqrt{k-i+1}}. \end{aligned}$$

Hence

$$|c_k| = \sum_{i=1}^k \frac{1}{\sqrt{i} \sqrt{k-i+1}}.$$

By the arithmetic-geometric mean inequality,

$$\sqrt{ab} \leq \frac{a+b}{2} \quad (a > 0, b > 0),$$

and with $a = i$, $b = k - i + 1$,

$$\frac{1}{\sqrt{i} \sqrt{k-i+1}} \geq \frac{2}{i+k-i+1} = \frac{2}{k+1}.$$

Therefore

$$|c_k| \geq \sum_{i=1}^k \frac{2}{k+1} = \frac{2k}{k+1} \geq 1.$$

Since $\lim_{k \rightarrow \infty} c_k \neq 0$, by [Claim 1.5](#) (the necessary condition for convergence), the series

$$\sum_{k=1}^{\infty} c_k$$

diverges.

The following two claims, which we give without proof, provide sufficient conditions for the implication above to hold.

Claim 4.1 (The Mertens Theorem). Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two convergent series with sums a and b , respectively. If at least one of them converges absolutely, then their Cauchy product ([Definition 4.1](#)) converges to ab .

Thus, the implication in the question is true if at least one of the series is absolutely convergent.

Claim 4.2 (The Abel Theorem on the Cauchy product). Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be two convergent series with sums a and b , respectively. If their Cauchy product

$$\sum_{k=1}^{\infty} c_k$$

(see [Definition 4.1](#)) is a convergent series, then

$$\sum_{k=1}^{\infty} c_k = ab.$$

Definition 4.2. Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two series. Let

$$\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2, \quad i \mapsto (m_i, n_i),$$

be a bijection from $\mathbb{N}$ to $\mathbb{N}^2$. We define the $\tilde{\tau}$ -product of the series $\sum a_k$ and $\sum b_k$ by

$$\left(\sum_{k=1}^{\infty} a_k \right) \dot{\underset{\tau}{\cdot}} \left(\sum_{k=1}^{\infty} b_k \right) = \sum_{i=1}^{\infty} a_{m_i} b_{n_i}.$$

Note 1. Since there are infinitely many bijections of the form $\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2$, there are infinitely many distinct definitions of the product of two series.

Note 2. The idea behind [Definition 4.2](#) is the following. If we need to multiply

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k,$$

we need to expand the brackets

$$(a_1 + a_2 + \cdots + a_k + \cdots)(b_1 + b_2 + \cdots + b_k + \cdots).$$

For example:
 “the serpent” order (= bijection $\tilde{\tau}$):



“the diagonal” order (= bijection $\hat{\tau}$):

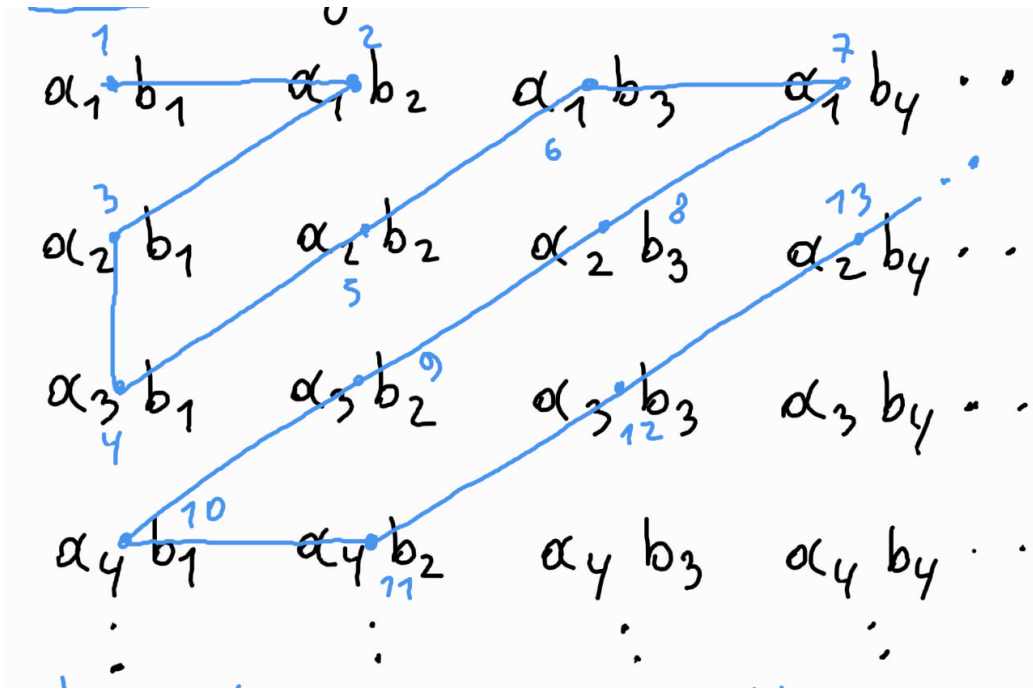


Figure 4.2

$$\left(\sum_{i=1}^{\infty} a_i \right) \cdot_{\tilde{\tau}} \left(\sum_{i=1}^{\infty} b_i \right) = a_1 b_1 + (a_1 b_2 + a_2 b_1) + (a_3 b_1 + a_2 b_2 + a_1 b_3) + (a_1 b_4 + a_2 b_3 + a_3 b_2 + a_4 b_1) + \cdots,$$

where terms c_i are the same as in the Cauchy product ([Definition 4.1](#)).

The next theorem shows that if both series are absolutely convergent, then the choice of the order does not matter.

Claim 4.3 (The Abel Theorem on Series Product). Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two absolutely convergent series with sums a and b , respectively. Then, for every bijection

$$\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2, \quad i \mapsto (m_i, n_i),$$

we have

$$\left(\sum_{k=1}^{\infty} a_k \right) \cdot_{\tilde{\tau}} \left(\sum_{k=1}^{\infty} b_k \right) = \sum_{i=1}^{\infty} a_{m_i} b_{n_i} = ab.$$

Proof. For every $N \in \mathbb{N}$, let

$$K = \max\{m_1, \dots, m_N, n_1, \dots, n_N\}.$$

Then

$$\sum_{i=1}^N |a_{m_i} b_{n_i}| \leq \left(\sum_{i=1}^K |a_i| \right) \left(\sum_{i=1}^K |b_i| \right).$$

The inequality holds because all terms are nonnegative and every term from the left-hand sum is present in the product on the right. Hence

$$\sum_{i=1}^N |a_{m_i} b_{n_i}| \leq d\beta,$$

where

$$\sum_{i=1}^{\infty} |a_i| = d, \quad \sum_{i=1}^{\infty} |b_i| = \beta.$$

Since the two series are absolutely convergent, $d, \beta \in \mathbb{R}$, and the sequences of partial sums

$$\left\{ \sum_{i=1}^n |a_i| \right\}_{n=1}^{\infty}, \quad \left\{ \sum_{i=1}^n |b_i| \right\}_{n=1}^{\infty}$$

are nondecreasing; in particular their partial sums are bounded by d and β , respectively.

Thus the sequence of partial sums

$$\left\{ \sum_{i=1}^N |a_{m_i} b_{n_i}| \right\}_{N=1}^{\infty}$$

is bounded by $d\beta$. Since this sequence is nondecreasing, [Claim 2.2](#) implies that

$$\sum_{i=1}^{\infty} |a_{m_i} b_{n_i}|$$

is convergent. By [Definition 3.1](#), the series

$$\sum_{i=1}^{\infty} a_{m_i} b_{n_i}$$

is absolutely convergent.

Therefore, by [Claim 3.9](#) (Cauchy's Rearrangement Theorem), no permutation $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ affects its sum. We may rearrange the series so that the order is diagonal:

$$a_1 b_1 + (a_1 b_2 + a_2 b_1) + (a_1 b_3 + a_2 b_2 + a_3 b_1) + \cdots + (a_1 b_k + a_2 b_{k-1} + \cdots + a_k b_1) + \cdots .$$

Grouping the terms gives

$$c_1 + c_2 + c_3 + \cdots + c_k + \cdots .$$

By [Definition 4.1](#), this is the Cauchy product of $\sum a_i$ and $\sum b_i$. Since these series are absolutely convergent, [Claim 4.1](#) (Mertens' theorem) gives

$$\sum_{k=1}^{\infty} c_k = ab.$$

Therefore

$$\sum_{i=1}^{\infty} a_{m_i} b_{n_i} = ab.$$

□

The Stirling Formula

To prove the formula, we will need the following fact.

Claim 4.4. Let

$$I_n = \int_0^{\pi/2} \sin^n(x) dx, \quad n \in \mathbb{N} \cup \{0\}.$$

Then:

- 1) the sequence $\{I_n\}_{n=1}^{\infty}$ is nonincreasing;

2) for every $n \in \mathbb{N}$,

$$I_{2n} = \frac{\pi}{2} \cdot \frac{(2n-1)!!}{(2n)!!}, \quad I_{2n+1} = \frac{(2n)!!}{(2n+1)!!}.$$

Proof. 1) For every $x \in [0, \pi/2]$,

$$0 \leq \sin x \leq 1.$$

Hence, for every $n \in \mathbb{N} \cup \{0\}$,

$$0 \leq \sin^{n+1} x \leq \sin^n x \leq 1,$$

and therefore

$$I_{n+1} = \int_0^{\pi/2} \sin^{n+1} x \, dx \leq \int_0^{\pi/2} \sin^n x \, dx = I_n.$$

Thus $\{I_n\}$ is nonincreasing.

2) We prove the equalities by induction on $n \in \mathbb{N} \cup \{0\}$.

For $n = 0$,

$$I_0 = \int_0^{\pi/2} \sin^0 x \, dx = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2} = \frac{\pi}{2} \cdot \frac{(2 \cdot 0 - 1)!!}{(2 \cdot 0)!!}.$$

For $n = 1$,

$$I_1 = \int_0^{\pi/2} \sin x \, dx = -\cos x \Big|_0^{\pi/2} = 1 = \frac{(2 \cdot 0)!!}{(2 \cdot 0 + 1)!!}.$$

For the induction step, note that

$$\begin{aligned} I_{n+2} &= \int_0^{\pi/2} \sin^{n+1} x \sin x \, dx \\ &= -\cos x \sin^{n+1} x \Big|_0^{\pi/2} + \int_0^{\pi/2} (n+1) \sin^n x \cos^2 x \, dx \\ &= (n+1) \int_0^{\pi/2} \sin^n x (1 - \sin^2 x) \, dx \\ &= (n+1) I_n - (n+1) I_{n+2}. \end{aligned}$$

Thus

$$I_{n+2} = \frac{n+1}{n+2} I_n \quad (n \in \mathbb{N} \cup \{0\}).$$

Therefore

$$I_{2n+2} = \frac{2n+1}{2n+2} I_{2n} = \frac{2n+1}{2n+2} \cdot \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} = \frac{\pi}{2} \frac{(2n+1)!!}{(2n+2)!!},$$

and

$$I_{2n+3} = \frac{2n+2}{2n+3} I_{2n+1} = \frac{2n+2}{2n+3} \cdot \frac{(2n)!!}{(2n+1)!!} = \frac{(2n+2)!!}{(2n+3)!!}.$$

□

Claim 4.5 (The Wallis Formula). Let

$$P_n = \prod_{k=1}^n \frac{(2k)^2}{(2k)^2 - 1} \quad (n \in \mathbb{N}).$$

Then

$$\lim_{n \rightarrow \infty} P_n = \frac{\pi}{2}.$$

Proof. For every $n \in \mathbb{N}$,

$$\frac{P_n}{P_{n-1}} = \frac{(2n)^2}{(2n)^2 - 1} > 1,$$

so the sequence $\{P_n\}_{n=1}^{\infty}$ is increasing.

Also,

$$\begin{aligned} P_n &= \prod_{k=1}^n \frac{(2k)^2}{(2k-1)(2k+1)} \\ &= \frac{(2 \cdot 4 \cdot 6 \cdots 2n)^2}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots (2n-1)(2n+1)} \\ &= \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1). \end{aligned} \tag{**}$$

By item 1) of Claim 4.4,

$$I_{2n+1} \leq I_{2n} \leq I_{2n-1}.$$

Using item 2) of Claim 4.4,

$$\frac{(2n)!!}{(2n+1)!!} \leq \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} \leq \frac{(2n-2)!!}{(2n-1)!!}.$$

Multiplying by $\frac{(2n)!!}{(2n-1)!!}$ gives

$$\left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1) \leq \frac{\pi}{2} \leq \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 \frac{(2n+1)^2}{2n}.$$

By (**),

$$P_n \leq \frac{\pi}{2} \leq P_n \frac{2n+1}{2n}.$$

Hence

$$\lim_{n \rightarrow \infty} P_n = \frac{\pi}{2}.$$

□

Claim 4.6 (Almost the Stirling Formula). There exists a limit

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{n} (n/e)^n} = A > 0.$$

That is, we prove that the limit exists and its value is positive.

Proof. Let

$$a_n = \frac{n!}{\sqrt{n} (n/e)^n} = \frac{n! e^n}{n^{n+1/2}} \quad (n \in \mathbb{N}).$$

We need to prove that $\lim_{n \rightarrow \infty} a_n = A > 0$.

Note that

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{(n+1)! e^{n+1}}{(n+1)^{n+1+1/2}} \cdot \frac{n^{n+1/2}}{n! e^n} \\ &= e \left(\frac{n}{n+1} \right)^{n+1/2} = e \left(1 + \frac{1}{n} \right)^{-n-1/2}. \end{aligned}$$

Hence

$$\begin{aligned}
\ln\left(\frac{a_{n+1}}{a_n}\right) &= 1 - \left(n + \frac{1}{2}\right) \ln\left(1 + \frac{1}{n}\right) \\
&= 1 - \left(n + \frac{1}{2}\right) \left(\frac{1}{n} - \frac{1}{2} \frac{1}{n^2} + \frac{1}{3} \frac{1}{n^3} + O\left(\frac{1}{n^4}\right)\right) \\
&= 1 - \left(1 - \frac{1}{2} \frac{1}{n} + \frac{1}{3} \frac{1}{n^2} + O\left(\frac{1}{n^3}\right)\right) - \left(\frac{1}{2} \frac{1}{n} - \frac{1}{4} \frac{1}{n^2} + \frac{1}{6} \frac{1}{n^3} + O\left(\frac{1}{n^4}\right)\right) \\
&= \left(-\frac{1}{3} + \frac{1}{4}\right) \frac{1}{n^2} + O\left(\frac{1}{n^3}\right) \\
&= \frac{1}{n^2} \left(-\frac{1}{12} + O\left(\frac{1}{n}\right)\right). \tag{1}
\end{aligned}$$

Let

$$P_n = \prod_{k=1}^n \frac{a_{k+1}}{a_k} = \frac{a_2}{a_1} \frac{a_3}{a_2} \dots \frac{a_{n+1}}{a_n} = \frac{a_{n+1}}{a_1}. \tag{3}$$

Then, by (1),

$$\ln(P_n) = \sum_{k=1}^n \ln\left(\frac{a_{k+1}}{a_k}\right) = \sum_{k=1}^n \frac{1}{k^2} \left(-\frac{1}{12} + O\left(\frac{1}{k}\right)\right). \tag{2}$$

Since the factor in parentheses is bounded for all sufficiently large k , the series converges absolutely by [Claim 2.4](#) and [Claim 2.7](#):

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \left(-\frac{1}{12} + O\left(\frac{1}{k}\right)\right)$$

converges, say to $b \in \mathbb{R}$. Therefore, from (2),

$$\lim_{n \rightarrow \infty} \ln(P_n) = b.$$

Thus

$$\lim_{n \rightarrow \infty} P_n = e^b \in \mathbb{R}. \tag{4}$$

Finally,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n+1} = a_1 \lim_{n \rightarrow \infty} P_n = a_1 e^b.$$

Since $a_1 = e$, we obtain

$$A = e^{b+1} > 0.$$

□

Claim 4.7 (The Stirling Formula).

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} (n/e)^n} = 1,$$

equivalently,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad n \rightarrow +\infty.$$

Note.

$$\lim_{n \rightarrow \infty} \left(n! - \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \right) = +\infty.$$

Proof. Due to [Claim 4.5](#) (the Wallis formula) and equality [\(**\)](#), we have

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1) = \frac{\pi}{2}.$$

Thus

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \sqrt{2n+1} \right) = \sqrt{\frac{\pi}{2}}. \quad (5)$$

On the other hand,

$$\frac{(2n)!!}{(2n+1)!!} = \frac{2 \cdot 4 \cdot 6 \cdots 2n}{1 \cdot 3 \cdot 5 \cdots (2n+1)} = \frac{2^{2n} (n!)^2}{(2n+1)!}.$$

Hence

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^{2n} (n!)^2}{(2n+1)!} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^{2n} (A\sqrt{n} (n/e)^n)^2}{A\sqrt{2n+1} ((2n+1)/e)^{2n+1}} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \frac{2^{2n} e A n^{2n+1}}{(2n+1)^{2n+1}} \\ &= \lim_{n \rightarrow \infty} \left(e A 2^{2n} \left(\frac{2n+1}{n} \right)^{-2n-1} \right) \\ &= \lim_{n \rightarrow \infty} \left(e A \cdot \frac{1}{2} \left(1 + \frac{1}{2n} \right)^{-2n} \left(1 + \frac{1}{2n} \right)^{-1} \right) \\ &= \frac{1}{2} A. \end{aligned}$$

Therefore, by [\(5\)](#),

$$\sqrt{\frac{\pi}{2}} = \frac{1}{2} A,$$

so

$$A = \sqrt{2\pi}.$$

By [Claim 4.6](#),

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi} \sqrt{n} (n/e)^n} = 1,$$

which is the Stirling formula. □

Claim 4.8 (The Robbins Formula). For every $n \in \mathbb{N}$, we have

$$\sqrt{2\pi n} \left(\frac{n}{e} \right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e} \right)^n e^{\frac{1}{12n}}.$$

Without proof.